

A. EDUCATION

PhD in Informatics (Focusing LLM)	Present
Postgraduate Diploma in Business Management Multimedia University	2022 – 2023
M.Sc., Information Technology Dissertation: Feature and Opinion Extraction on Malay Online Text	2015 – 2019
B.Sc., Bioinformatics	2011 – 2014

B. Certification

Digital Maturity Disruptor	2024
SLB-CI White Belt	2024
Dataiku ML Practitioner	2023
Dataiku-SLB- Core Designer	2023
Oxford Machine Learning (Summer School)	2023
Data Science & ML (MIT Summer School)	2022
Certified Big Data Analytics Practitioner (ACE)	2020
Certificate: Certified Data Science Specialist (iTrain)	2020
The Data Science Course 2020: Complete Data Science Bootcamp	2018

C. WORK EXPERIENCE

Machine Learning Research Officer – (i) (MMU: Telekom Sentiment Project)	Oct 2014- Apr 2018
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- Develop statistical learning models and NLP systems tailored to organizational needs, focusing on text mining and data processing.
- Collaborate with various departments to understand company requirements and create effective solutions, while staying updated on technology trends.
- Implement and enhance mathematical and statistical methodologies for analytical models, driving decision-making with insights from data analysis.
- Utilize and extend machine learning libraries for NLP tasks, ensuring the selection of appropriate algorithms and tools.
- Manage data collection, processing, and validation, employing tools and processes for extracting information from unstructured data.

- Conduct training, evaluation, and optimization of models, collaborating with researchers for validation and fine-tuning through experiments.
- Maintain responsibility for project timelines, effectively communicating progress and results to team leaders and stakeholders.

Impact:

- Create the pioneering digital lexicon for the Malay language, featuring an automatic update function for adding new words.
- Assist the organization's R&D team in developing methods to accurately identify opinion words within Malay sentences.
- Developed a feature-based sentiment analysis system for a major Malaysian telecommunications provider, analyzing over 10,000 customer reviews monthly and saving approximately 1 million MYR in third-party service fees.

(ii). Lecturer - Limkokwing University (Student Coordinator)

Oct 2017-Dec 2018

- Conducting teaching and research in Artificial Intelligence and Business Intelligence.
- Managing the administration of courses, organizing lectures, seminars, and tutorials.
- Designing new courses and materials and assessing students' work.
- Overseeing exams, marking results, and performing departmental administrative tasks.
- Providing personal mentoring and support to students and implementing university research projects.
- Preparing reports for the management team, leading discussions, and promoting student engagement.
- Participating in student recruitment, education fairs, and managing events for international students.

Impact:

- Designing new courses and materials ensures that the curriculum remains cutting-edge and relevant, preparing students for future challenges and opportunities.
- Implemented innovative teaching methodologies and research projects that led to a 20% increase in student engagement and a 15% improvement in academic performance in AI-related modules across 250 students.

(iii). **Data Scientist - Sapura Energy Berhad**

Jan 2019 – Jan 2021

- Leading or supporting 1 to 4 projects in Business Intelligence (BI) & Analytics, engaging with global stakeholders to translate business needs into BI/Analytics solutions.
- Utilizing analytical models, optimization models, forecasting algorithms, and AI models to drive key decisions and meet business needs.
- Leveraging tools like Python, PowerBI, Excel, Spark, and Scikit-learn for efficient analytics, including data-driven optimizations, experiments, and forecasting.
- Working cross-functionally to define problem statements, collect data, build analytical models, and make recommendations, including governing material and vendor master data in SAP and Symphony.
- Applying statistical and analytical techniques for data analysis and reporting, including web scraping and sentiment analysis on social media data.
- Building real-time reporting systems connecting SAP, Business Warehouse, PowerBI, and SharePoint, including automating workflows and calculations.
- Supporting digital transformation by analyzing and modeling data, implementing advanced statistical methods, machine learning techniques, and deep learning methods for new material hierarchy mapping.

Impact:

- Implemented advanced deep learning with word embedding to reduce material mapping efforts from 3 days to 15 minutes.
- Developed dashboards for anomaly detection in material catalogs to reduce incorrect material group selection by buyers from 40% to 5%, impacting over 400 procurement transaction monthly.
- Introduce unique vendor and material/service codes which increase buyer efficiency by 90%.
- Streamlined financial reporting from one week to minutes using SAP, SAP BW, and PowerBI.
- Introduced ML for Vendor Classification to boost reliability and cut manual effort on risk assessment.
- Led Data Science awareness training across departments, promoting data-driven decision-making.

(iv). **Data Scientist - Natural Intelligence Solutions Sdn. Bhd.**

Jan 2021 – Jan 2022

- Perform comprehensive data analysis, including data preparation, modeling, and interpretation to develop custom data models and algorithms.
- Utilize predictive modeling to assist clients in enhancing business processes, revenue forecasting, and cost reduction.
Design and implement various data science studies (descriptive, predictive, and prescriptive) to generate business insights and actionable strategies.

- Participate in client analytics projects, including segmentation, churn prediction, forecasting, and optimization.
- Support senior team members through workshops/interviews, contribute to business development, and collaborate with multiple teams for solution deployment.
- Aid in managing analytics infrastructure, design technical solution prototypes, and ensure high-quality output from the Data Science team.
- Provide training on PowerBI, statistics, and Python.

Impact:

- Slashed month-long manual reporting to 2 days with an automated data pipeline for consolidated reporting across 56 child companies.
- Aid in designing advance analytic dashboard which includes product demand forecasting and secured about 1 million MYR project.
- Drive a 20% increase in efficiency through the design and implementation of digital transformation initiatives for HP organizations.

(v). **Data Scientist - AINQA Healthcare**

Jan 2022 – Aug 2022

- Evaluate the effectiveness and accuracy of new data sources and collection methods, ensuring they are ethical and efficient.
- Develop and tailor custom data models and algorithms for specific project goals, ensuring they are deployable in production.
- Collaborate with various teams for model implementation, monitoring outcomes, and clear communication to prevent misinterpretation.
- Monitor, analyse, and explain model performance and data accuracy to stakeholders, ensuring results are meaningful and broadly accepted.
- Handle ad hoc projects and tasks as directed by supervisors.

Impact:

- Spearheaded the development of ad hoc models for UAE COVID-19 forecasting, playing a critical role in securing a million MYR deal.
- Successfully collaborated with Malaysian public health professionals to develop a data-driven model for forecasting TB incidents.
- This model utilized parametric and non-parametric analyses of meteorological, air quality, festival, and population density data, potentially improving public health preparedness.

- Responsible for communicating analytics model. behavior/results to business.
- Responsible for carrying out technical risk analysis and reliability assessments.
- Responsible for performing complex analyses, including optimization, text analytics, machine learning, social-science modeling, and statistical analysis, parametric and non-parametric statistical models and techniques.
- Responsible for the design of algorithms that require different models/methods to be used together.
- Apply strong understanding of data science techniques and libraries to business problems.
- Direct data science teams to complete all project deliverables.
- Provide direction and mentorship for the GBS Data Science focus group and foster a culture of data & analytics in the GBS functional teams.

Impact:

- Leveraged advanced text analytics to identify and automate non-value-added activities (e.g., cancelling incidents), saving analysts 26 hours/month.
- Developed predictive models for incident classification, resulting in 4,050 hours/year saved and reducing operational costs by ~\$60,750 USD/year.
- Built an ensemble model for efficient budgeting allocation, streamlining geounit champion workload.
- Implemented a Retrieval-Augmented Generation (RAG) model utilizing LLMA2 for desk service, achieving a significant 30% reduction in incident volume.

D. VOLUNTEER PROJECTS**i. Modeling economic well-being in India with the World Resources Institute**

- As a Machine Learning Engineer, enrolled a two-month collaborative project with the World Resources Institute focusing on modeling economic well-being in India. This initiative involved **teaming** up with a global team of up to 50 innovators to develop machine learning models utilizing publicly available satellite imagery and ground data to enhance understanding of economic conditions. The key responsibilities in this project included data acquisition, feature engineering using NDVI and AWI, insight extraction through analytical modeling and the training of models to detect urbanization and agricultural changes indicative of economic well-being.

ii. **Analyzing the Role of Connectivity on Economic & Human Development with (United Nation Development Plan) UNDP.**

- As a Lead Machine Learning Engineer in a critical project with the United Nations Development Programme (UNDP) titled "Analysing the Role of Connectivity on Economic & Human Development." Key responsibilities included acquiring and preparing datasets through comprehensive feature engineering, conducting extensive Exploratory Data Analysis (EDA) to derive meaningful insights using both statistical and non-statistical methods. Conduct meeting and discussion on experimenting machine learning models to predict human development index (HDI). This effort highlighted the use of advanced analytics to influence global policy and development strategies.

iii. **Reinforcement ML model with Business Love Intelligent**

- Led the development of an innovative Reinforcement Machine Learning model for the Business Love Intelligent (BLI) tool, aimed at enhancing audience engagement strategies. This project involved designing a sophisticated ML model capable of predicting Audience Cluster Receptivity Score (%) and Brand Receptivity Score (%) together with the mapping of the receptivity stage of global audience clusters. A key component was the creation of a working personality model that accurately forecasts Big 5 personality traits and headed a team in developing and experimenting sentiment mining models.

iv. **Helping People with Visual Impairment to Easily Use Buses through Computer Vision**

- Collaborated with a fast-paced team of 35 to develop an automated computer vision pipeline within two months, improving the quality of life for visually impaired individuals. Played a key role by annotating data in CVAT and experimenting with the YOLOv5 model for empty seat detection.

v. **Predicting Cardiac Arrest Using AI**

- Joined development team of an AI-based solution for predicting cardiac arrest which focuses on comprehensive data handling and advanced machine learning techniques. The responsibilities included conducting exploratory data analysis to understand data characteristics and inform methodology selection, acquiring and cleansing data for consistency. It is also involved in implementing rigorous feature engineering to identify key predictors. Developed and evaluated predictive models and employing a variety of machine learning algorithms to optimize performance and accuracy in forecasting cardiac arrest incidents.

vi. **Facilitating Access to Financial Services for Underserved Populations in Africa**

- As a Lead Machine Learning Engineer, joined a transformative project to improve financial access for Africa's underserved populations. To uncover key insights, extensive Exploratory Data Analysis (EDA) is conducted on financial datasets provided by Toju, Africa. Led feature engineering and models experimenting team to design and improve the prediction model accuracy.

E. SKILLS

- a) Programming languages: Python, R language, SQL, C#
- b) Machine Learning/ Data Science tools: TensorFlow, Pytorch, Scikit-Learn, Pandas, Numpy
- c) Cloud and Deployment Tools: Dataiku, Alteryx, ADF, web PowerBI, ServiceNow, Databricks, SharePoint, PowerApps, SAP, AWS
- d) Data Mining Techniques: Machine Learning, Deep Learning, Natural Language Processing, LLM
- e) Database: MySQL
- f) Data Visualization: Microstrategy, PowerBI, Tableau, Streamlit

F. DATA COMMUNITY

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| 1. Judge for <i>Kementerian Pelajaran Malaysian</i> (KPM) STEM Competition. | 2019 |
| 2. Speaker - Omdena: <i>“AI in Agriculture”</i> | 2020 |
| 3. Speaker - IE University: <i>“Consumer Analytics”</i> | 2020 |
| 4. Speaker - Mahsa University: <i>“The Transfer and Diffusion of Information Technology for Organization”</i> | 2020 |
| 5. Panel Guest - Omdena: <i>“Improving Food Security & Crop Yield in Malaysia Using ML”</i> | 2021 |
| 6. Judge - Science Film Festival: <i>“Virtual Quiz”</i> | 2021 |
| 7. Speaker - University Technology Petronas: <i>“Python Programming”</i> | 2021 |
| 8. Speaker - Big Data Days: <i>“AI is The New Electricity”</i> | 2022 |
| 9. Malaysia AI Trainer: <i>“Recommendation System”</i> | 2022 |
| 10. Speaker – EAGE Workshop on Data Science. <i>“Service Desk Analytics: Enhancing User Experiences and Efficiency”</i> | 2023 |

G. RESEARCH INTEREST

- Sentiment Mining
- Personalized Information Retrieval
- Machine Learning
- Deep Learning
- Natural Language Processing

H. JOURNAL PUBLICATIONS

- i. S. M. A. Kalaiarasi, S. Surendran, and K. K. Jaya, "The Generation of Malay Lexicon" American Journal of Applied Sciences, vol. 14, pp. 503-510, 2017.
- ii. Selvaraju, S., & Anbananthen, K. S. M. (2019). Opinion Extraction on Online Malay Text. American Journal of Applied Sciences, 16(4), 134-142.
- iii. Selvaraju, S., Sjarif, N. N. A., & Anuar, M. S. M. (2024, September). Sentiment Analysis in Code Mixing Malay Text: A Review. In 2024 5th International Conference on Artificial Intelligence and Data Sciences (AiDAS) (pp. 280-285). IEEE.
- iv. Selvaraju, S., Sjarif, N. N. A., & Anuar, M. S. M. (2024, September). Code Mixing Malay Online Text Sentiment Analysis: Proposed Feature Extraction and Feature Selection Scheme. In 2024 5th International Conference on Artificial Intelligence and Data Sciences (AiDAS) (pp. 1-6). IEEE.
- v. "Driving AI Democratization: A Strategic Framework for Wide-Scale Adoption Across Organization "EAGE Workshop on Data Science - From Fundamentals to Opportunities, 2023 V. Mirje; Y.F. Lim; C.E. Dragomir; Y. Shariffuddin; S. Selvaraju
- vi. "Unleashing the Power of Text Analytics: Enhancing Service Desk Efficiency and User Satisfaction" EAGE Workshop on Data Science - From Fundamentals to Opportunities, 2023 S. S Selvaraju

I. LINKS

- i. **GitHub** - <https://github.com/Sureniology>
- ii. **LinkedIn** - <https://www.linkedin.com/in/surendran-selvaraju-b8040185/>

J. AWARDS

- **Gold medal ITEX'16**
- **Gold medal MMU INOVATION'16**
- **Silver medal PENCIPTA' 15**
- **Youth Icon MIYC' 22**

K. REFERENCES

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